An Empirical Analysis of ChatGPT's Impact on Developer Productivity Using Italy's ChatGPT Ban as a Natural Experiment

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CCS Concepts: • **Software and its engineering** → **Empirical software engineering**; *Software development process management*; • **Human-centered computing** → *Empirical studies in HCI*; • **Computing and society** → Impact of computing on society; • **Computing methodologies** → Natural language processing.

Additional Key Words and Phrases: ChatGPT, generative AI, developer productivity, software engineering, empirical study, GitHub, policy impact, Italy

ACM Reference Format:

Richard Schulz and Simon Schuhmacher. 2018. An Empirical Analysis of ChatGPT's Impact on Developer Productivity Using Italy's ChatGPT Ban as a Natural Experiment. In *Proceedings of Make sure to enter the correct conference title from your rights confirmation email (Conference acronym 'XX)*. ACM, New York, NY, USA, 11 pages. https://doi.org/XXXXXXX.XXXXXXX

1 Introduction

Substantial anecdotal evidence suggests that AI-assisted coding tools such as ChatGPT[1] have significantly increased developer productivity. However, recent empirical research has shown mixed results [3, p. 12], [8, p. 2], [9, p. 5] [2, p. 2].

Despite this growing literature, much of the existing evidence is correlational or based on controlled experiments detached from real-world settings. We still know little about how sudden changes in access to such tools affect developers in practice. The short-lived ChatGPT ban in Italy in early 2023 [6] offers a rare quasi-experimental opportunity to observe developers' behavioral responses to an abrupt disruption in access to a key AI tool.

By analyzing GitHub repositories, Kreitmeir and Raschky [7, p. 7] documented a 50% decline in release events in Italian open source projects during the first days of the ban. Building on these findings, we investigate the effect of the ChatGPT ban on other developer output, particularly on added and deleted lines of code and the number of pull requests created. These variables reflect developer activity, one of the key dimensions of the SPACE framework used to measure developer productivity [4, pp. 49–50]. Based on the findings of previous research, which shows using AI-assisted coding tools reduced task completion time by over 55% [9, p. 5], we assume the daily volume of code contributed by developers is higher when they have access to such tools. Similarly, the number of pull requests has also been employed in related research as a proxy for developer productivity, where significant increases have been observed following the

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[1] https://chatgpt.com/overview/

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Manuscript submitted to ACM

introduction of AI-assisted coding tools [3, pp. 6, 12]. Therefore, by blocking access to such tools, we would expect a decrease in the corresponding variables as a reaction to the ChatGPT ban.

To investigate how the ChatGPT ban affected developers' coding activity, we address the following research questions:

- **RQ1**: Did the ChatGPT ban cause a reduction in the daily volume of code contributed by developers in Italy, as measured through lines of code added and deleted per user per day?
- **RQ2**: Did the ChatGPT ban lead to a decrease in the number of pull requests opened by developers in Italy, indicating a reduction in feature-level output?

These research questions allow us to examine the immediate behavioral response to the ban, specifically coding activity and feature-level output. By focusing on granular metrics, we aim to provide a more comprehensive understanding of how the disruption in AI tool access affected the day-to-day development process.

2 Related Work

AI-assisted coding tools have introduced substantial changes to how developers create and maintain software. This section reviews prior research in three areas relevant to our study. We first discuss work examining the effects of AI-assisted coding tools on developer productivity. We then consider research that leveraged the temporary ChatGPT ban in Italy in 2023 as a natural experiment. Finally, we discuss how developer productivity can be measured using previously collected data.

2.1 AI-Assisted Coding Tools and Developer Productivity

A growing body of empirical research has investigated the impact of AI-assisted coding tools on developer productivity, with mixed results. Early studies on GitHub Copilot found significant productivity gains, with developers completing tasks faster when using AI code completion [9, p. 5]. More recent large-scale field experiments have confirmed productivity improvements in some contexts. Cui et al. [3, pp. 12, 19] conducted randomized controlled trials across three companies and found a 26% increase in pull requests created among developers using AI tools, with less experienced developers showing greater gains. Paradis et al. [8, p. 2] found that productivity improvements in enterprise settings vary substantially by task complexity and developer experience.

However, not all studies show positive effects. Becker et al. [2, p. 2] conducted a randomized controlled trial with experienced open-source developers and found that those using AI coding tools took 19% longer to complete tasks in familiar codebases compared to developers not using AI tools. The authors attributed this slowdown to the time spent reviewing and correcting AI-generated code, suggesting that experienced developers may face verification overhead that outweighs the benefits of AI assistance in contexts where they already possess strong domain knowledge. This finding highlights that AI tools' impact on productivity is highly context-dependent and may not universally benefit all developer populations.

Existing studies largely rely on controlled experiments or correlational data, leaving open how developers react to the sudden, unanticipated loss of AI tools. Most research focuses on adoption effects rather than disruption responses.

2.2 The ChatGPT Ban as a Natural Experiment

Kreitmeir and Raschky [7, p. 7] were the first to leverage Italy's ChatGPT ban as a quasi-experimental setting to study the real-world impact of AI tool access. Using a difference-in-differences design, they analyzed GitHub release events and documented a 50% decline in releases from Italian open-source projects during the first days of the ban. This work established the ban as a valuable natural experiment and demonstrated the feasibility of using GitHub activity data to study developer behavior at scale.

However, their analysis focused on aggregate project-level outcomes (release events), which capture only high-level project milestones. A release may bundle multiple features and code changes, making it less sensitive to immediate shifts in day-to-day coding activity. Additionally, releases might be influenced by factors beyond individual developer productivity, such as project management practices and team coordination. The granular, day-to-day development process—including actual code production indicators—remains unexamined. Our work extends this foundation by investigating code-level and feature-level metrics that provide more direct measures of developer activity.

2.3 Measuring Developer Productivity

Prior research has demonstrated that developer productivity is influenced by multiple factors and, therefore, cannot be accurately captured by a single metric [4, pp. 47–48]. Moreover, measuring output quantity—such as the number of lines of code a developer adds—can encourage behavior aimed at optimizing the metric rather than achieving meaningful progress [11, p. 60]. However, when working with retrospective data, alternative metrics are often unavailable. In such cases, and particularly when developers are unaware that their output is being measured, the risk of such gaming is relatively low. Accordingly, in this study, we use activity measures (lines of code added & deleted and number of pull requests opened) as proxies for developer productivity, while acknowledging the inherent limitations of this approach.

3 Methodology

To examine the impact of Italy's temporary ChatGPT ban on developer productivity, we employ a difference-in-differences research design using GitHub activity data, comparing Italian developers with their counterparts in Austria and France.

3.1 Data

The empirical analysis uses retrospective GitHub activity data from users in Italy, Austria, and France for the weeks surrounding the early April 2023 ban. We leverage the GH Archive[2] dataset, which provides a comprehensive record of public GitHub events. Since GH Archive data does not include developers' location information, we infer users' countries based on their public profiles, following the methods from prior research [7, p. 5]. We analyze `PushEvents` emitted by developers identified within our treatment and control groups. For each event, we retrieve the corresponding commit data through the GitHub REST API,[3] extracting the number of lines added and deleted. We then compute the average daily change in code volume per developer. Additionally, we extract `PullRequestEvents` from GH Archive with the attribute `action = 'opened'` to compute the average daily number of pull requests opened per developer in our sample.

3.2 Control Group Selection

We follow Kreitmeir and Raschky [7, p. 4] in selecting Austria and France as the control group for our difference-in-differences analysis. This choice is motivated by several factors that strengthen the validity of the parallel trends' assumption, which is critical for causal inference in DiD designs [1, p. 171]. First, Austria and France share cultural and linguistic similarities with Italy, being neighboring European countries with comparable levels of economic

[2]https://www.gharchive.org/
[3]https://docs.github.com/en/rest

development and technology adoption. Second, these countries were not affected by the ChatGPT ban, making them suitable counterfactuals for what would have happened in Italy in the absence of the ban. This selection helps ensure that any observed differences between Italy and the control countries during the ban period are attributable to the ban itself rather than to pre-existing differences in development trends.

3.3 Research Design

We employ a **Difference-in-Differences (DiD)** quasi-experimental design to ensure causal inference, replicating the framework used by Kreitmeir and Raschky [7, p. 6]. The DiD approach combines a before-after comparison with a treatment-control group design, enabling us to identify the impact of Italy's ChatGPT ban while accounting for time-varying factors affecting all countries [5, pp. 520–522]. The analysis compares the change in developer metrics in Italy (treatment group) against those in Austria and France (control group).

The ban was implemented on April 1, 2023. Our analysis focuses on a time window spanning four weeks before the ban (March 4–31, 2023) and two weeks after the ban (April 1–14, 2023). This time window allows us to capture both the immediate impact of the ban and potential adaptation effects. Additionally, following Kreitmeir and Raschky [7, p. 7], we examine the first four working days after the ban (April 3–6, 2023) separately to assess the immediate impact before circumvention behavior (such as VPN usage, as documented in Appendix B) may have mitigated the ban's effects over time. This design allows us to address the research questions (RQ1 and RQ2) by examining whether the ban caused statistically significant changes in code volume and the number of created pull requests.

In contrast to the original study—which measured aggregate productivity through project releases—our analysis zooms in on code-level and issue-level metrics that capture the day-to-day development process. The main dependent variables are:

- **Code Volume**: The number of lines of code (LOC) added and deleted per developer per day. This provides a granular measure of active coding behavior.
- **Pull Requests Opened**: The number of pull requests opened per developer per day, serving as a proxy for feature-level output.

We use the DiD specification represented in Equation 1 for our analyses.

$$Y_{it} = \beta D_{it} + \alpha_i + \lambda_t + \sigma_d \times t + \varepsilon_{it} \quad (1)$$

All variables will be standardized at the developer-day level to account for individual differences in coding activity. Y_{it} denotes the outcome variable per user i at time t. β (Italy × Post) is the parameter of interest, describing the DiD estimate. D_{it} equals to 1 for events made by users in the treatment group (Italy) during the post ban period that was analyzed (two weeks or four weekdays). We include user and date fixed effects in our regression specifications to control for unobserved heterogeneity across developers and time-invariant day-specific factors that may confound the treatment effect. User fixed effects (α_i) account for individual developer characteristics (e.g., skill level, coding style, project preferences) that remain constant over time, while date fixed effects (λ_t) capture common temporal shocks affecting all developers on a given day (e.g., holidays, major technology events, or platform-wide changes). We also include an interaction term between Italy and a linear time trend (Italy × Trend), represented as $\sigma_d \times t$ in the equation, to account for potential differential pre-treatment trends between Italy and the control countries, though its coefficient is not reported in the result tables as it is not of primary interest. Standard errors ε_{it} are clustered on the user-level.

To explore whether the ban's effects varied across different types of developers, we conduct a heterogeneity analysis by stratifying developers into activity groups based on pre-ban commit volume tertiles. We focus on activity level for several theoretically motivated reasons. First, prior research suggests that AI tool adoption and benefits vary by developer experience and engagement [3, p. 19], with less experienced developers showing greater productivity gains from AI assistance while experienced developers may face verification overhead. Activity level serves as a proxy for both developer engagement and potentially for reliance on AI tools—more active developers may have integrated AI tools more deeply into their workflows, making them more vulnerable to disruption. This analysis allows us to test whether the ban's impact varied systematically by baseline coding activity, which could reveal differential circumvention patterns or varied dependence on AI assistance across developer types.

4 Results

4.1 Code Volume Analysis

To address RQ1, we examine whether the ChatGPT ban affected developers' coding activity, measured by lines of code (LOC) added and deleted per developer per day. Table 1 presents the difference-in-differences estimates for two post-ban periods: the two weeks following implementation (April 1–14, 2023) and the first four working days (April 3–6, 2023).

Table 1. Difference-in-Differences Results: LOC Added / Deleted

	Log Additions		**Log Deletions**	
	2 Weeks	4 Weekdays	2 Weeks	4 Weekdays
Italy × Post	0.0177	0.0542	0.0136	0.0326
	(0.0486)	(0.0527)	(0.0489)	(0.0524)
Observations	224, 776	185, 999	224, 776	185, 999
R^2	0.428	0.449	0.425	0.444

Notes: Absorbed fixed effects for users and dates. Robust standard errors are clustered on the user-level in parentheses. *** p<0.01, ** p<0.05, * p<0.1

The treatment effect ("Italy × Post-Ban") is small and statistically insignificant in all specifications. For log additions, coefficients are 0.0177 (SE: 0.0486) for the two-week period and 0.0542 (SE: 0.0527) for the four-day period; for log deletions, 0.0136 (SE: 0.0489) and 0.0326 (SE: 0.0524), respectively. Overall, the ban did not measurably affect coding volume.

4.1.1 *Heterogeneity Analysis.*

To examine heterogeneity, we stratify developers into low-, mid-, and high-activity groups based on pre-ban commit tertiles. Table 2 presents separate estimates for each group.

Table 2. Heterogeneity Analysis by Developer Activity Level

	Log Additions			**Log Deletions**		
	Low	Mid	High	Low	Mid	High
Italy × Post	0.0238	0.0741	0.0016	-0.0908	-0.0196	0.0192
	(0.2172)	(0.1085)	(0.0560)	(0.2145)	(0.1068)	(0.0565)
Observations	17,424	42,697	157,932	17,424	42,697	157,932

Notes: Activity groups defined by pre-ban commit volume tertiles. Absorbed fixed effects for users and dates. Robust standard errors are clustered on the user-level in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Estimated effects remain small and statistically insignificant across all activity levels, indicating no evidence of heterogeneous responses to the ban.

4.1.2 Ban Lift Analysis. We next examine the period after the ban was lifted on April 28, 2023 (see Appendix A). This tests whether coding activity rebounded once formal access to ChatGPT was restored.

Across both the two-week and four-day post-lift windows, estimated treatment effects for LOC added and deleted remain small and statistically insignificant. The absence of any rebound implies that developers had already regained access—likely through circumvention methods—before the official lift. Combined with the null results during the ban, this suggests that the policy's practical impact on coding productivity was minimal.

4.2 Pull Requests Opened Analysis

To address RQ2, we assess whether the ban affected the number of pull requests opened per developer. Table 3 reports the difference-in-differences estimates for the same time windows as in the code volume analysis.

Table 3. Difference-in-Differences Results: Pull Requests Opened

	Log Pull Requests Opened	
	2 Weeks	4 Weekdays
Italy × Post	0.0175	0.0306^{*}
	(0.0162)	(0.0181)
Observations	47,943	39,794
R^2	0.387	0.402

Notes: Absorbed fixed effects for users and dates. Robust standard errors are clustered on the user-level in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Similar to the results from the code volume analysis, the results show no statistically significant effects on pull requests opened in the two-week period. For the four-weekday period, we observe a treatment effect of 0.0326 (SE: 0.0524), which is statistically significant at the 10% level ($p = 0.0897$).

5 Discussion

Our difference-in-differences analysis found no statistically significant effects on coding activity (RQ1) or pull request creation (RQ2) during the ChatGPT ban. Given field experiments showing productivity gains from AI adoption [9, p. 5], [3, p. 12], one might expect reduced coding output when access is disrupted. However, we observe no such reductions. This null result should not be interpreted as evidence that generative AI lacks utility, but rather as evidence of the high *resilience* and *adaptability* of developer workflows when faced with external disruptions.

The null findings for lines of code added and deleted are likely explained by widespread ban circumvention through VPN usage and other methods. As documented in Appendix B, VPN searches in Italy increased by 28.6 points (on a 0–100 scale) more than in control countries following the ban implementation, an effect that is statistically significant at the 1% level. This substantial and immediate spike—reaching peak popularity (index 100) on the very day the ban was implemented (see Figure 1)—suggests that many developers quickly regained access to ChatGPT despite the ban, effectively nullifying the treatment. While Google Trends data aggregates interest across the general population, software developers possess the specific *technological fluency* required to rapidly adopt circumvention tools. If developers circumvented the ban and continued using ChatGPT, we would expect to observe no reduction in coding activity—which is precisely what we find.

Several pieces of evidence support this circumvention interpretation. First, the heterogeneity analysis found no evidence of differential effects across developer activity groups (low, mid, and high activity), suggesting that the ban did not have statistically detectable impacts on coding activity regardless of developers' baseline activity levels. This uniformity across developer types is consistent with widespread circumvention rather than the genuine absence of treatment effects. Second, the ban lift analysis further corroborates this interpretation: **if there was no initial dip in coding activity due to circumvention, there can be no rebound when the ban is lifted**. By the time the ban was officially lifted on April 28, many developers had already circumvented the restriction weeks earlier, meaning the lift had no meaningful impact on their access. The null results in both the ban period and the ban lift period are therefore mutually reinforcing.

The stability of the Lines of Code (LOC) metric suggests that developers successfully maintained their AI-augmented workflows by bypassing the geographic restriction, effectively treating the ban as a technical hurdle rather than a hard constraint. This behavioral response demonstrates a key human aspect of software engineering: when developers perceive a tool as valuable to their workflow, they will actively seek ways to maintain access, even in the face of legal and technical barriers. Consequently, the "treatment" in this natural experiment was likely porous.

Additional factors may have contributed to the resilience of developer workflows. Although the ChatGPT website was blocked, access to the OpenAI API remained possible [10], and AI tools integrated into development environments, such as GitHub Copilot, were unaffected. These alternative pathways likely sustained AI access. It should also be considered that ChatGPT was still relatively new at the time of the ban (April 2023), which might have limited its integration into developers' workflows. As developers more deeply integrate AI tools into daily practice, future disruptions could have more pronounced effects—or conversely, could trigger even more determined circumvention efforts.

For pull requests opened, we observed a small positive treatment effect (0.0326, SE: 0.0524) during the four-weekday period, significant at the 10% level. However, given the small effect size, limited statistical significance, and absence of effects in the longer two-week period, this result may reflect random variation rather than a true treatment effect. The overall pattern for pull requests is consistent with the coding volume findings, suggesting no meaningful impact of the ban on feature-level output.

6 Conclusion

This study examined the impact of Italy's April 2023 ChatGPT ban on developer productivity using a difference-in-differences approach comparing Italian developers with peers in Austria and France. We found no significant decline in code volume or feature-level output during the ban.

However, this null effect reflects widespread circumvention rather than the irrelevance of AI tools. VPN search interest in Italy rose by 28.6 points (significant at the 1% level) immediately after the ban, and the absence of a post-lift rebound suggests most developers had already restored access.

These results highlight developers' strong reliance on AI-assisted workflows and their willingness to overcome barriers to maintain them. The rapid uptake of circumvention illustrates both technical fluency and a deep behavioral commitment to AI integration. As such tools become further embedded in software development, future disruptions are likely to prompt similar adaptive responses.

References

[1] Joshua D. Angrist and Jörn-Steffen Pischke. 2009. *Mostly Harmless Econometrics: An Empiricist's Companion.* Princeton University Press. http://www.jstor.org/stable/j.ctvcm4j72

[2] Joel Becker, Nate Rush, Elizabeth Barnes, and David Rein. 2025. Measuring the Impact of Early-2025 AI on Experienced Open-Source Developer Productivity. arXiv:2507.09089 [cs.AI] https://arxiv.org/abs/2507.09089

[3] Zheyuan Cui, Mert Demirer, Sonia Jaffe, Leon Musolff, Sida Peng, and Tobias Salz. 2025. The Effects of Generative AI on High-Skilled Work: Evidence from Three Field Experiments with Software Developers. http://dx.doi.org/10.2139/ssrn.4945566

[4] Nicole Forsgren, Margaret-Anne Storey, Chandra Maddila, Thomas Zimmermann, Brian Houck, and Jenna Butler. 2021. The SPACE of developer productivity. *Commun. ACM* 64, 6 (May 2021), 46–53. doi:10.1145/3453928

[5] Anders Fredriksson and Gustavo Magalhães de Oliveira. 2019. Impact evaluation using Difference-in-Differences. *RAUSP Management Journal* 54, 4 (Sept. 2019), 519–532. doi:10.1108/RAUSP-05-2019-0112 _eprint: https://www.emerald.com/rausp/article-pdf/54/4/519/2278699/rausp-05-2019-0112.pdf.

[6] Garante per la Protezione dei Dati Personali. 2023. Intelligenza artificiale: il Garante blocca ChatGPT. Raccolta illecita di dati personali. Assenza di sistemi per la verifica dell'età dei minori. Official Press Release. https://www.garanteprivacy.it/home/docweb/-/docweb-display/docweb/9870847 Published: 2023-03-31. Accessed: 2025-10-22.

[7] David H. Kreitmeir and Paul A. Raschky. 2023. The Unintended Consequences of Censoring Digital Technology – Evidence from Italy's ChatGPT Ban. arXiv:2304.09339 [econ.GN] https://arxiv.org/abs/2304.09339

[8] Elise Paradis, Kate Grey, Quinn Madison, Daye Nam, Andrew Macvean, Vahid Meimand, Nan Zhang, Ben Ferrari-Church, and Satish Chandra. 2024. How much does AI impact development speed? An enterprise-based randomized controlled trial. arXiv:2410.12944 [cs.SE] https://arxiv.org/abs/2410.12944

[9] Sida Peng, Eirini Kalliamvakou, Peter Cihon, and Mert Demirer. 2023. The Impact of AI on Developer Productivity: Evidence from GitHub Copilot. arXiv:2302.06590 [cs.SE] https://arxiv.org/abs/2302.06590

[10] Michele Saba. 2023. ChatGPT Bloccato in Italia: Che Fare? https://www.programmareinpython.it/blog/chatgpt-bloccato-in-italia-che-fare/ Accessed: 2025-11-24.

[11] Albert Ziegler, Eirini Kalliamvakou, X. Alice Li, Andrew Rice, Devon Rifkin, Shawn Simister, Ganesh Sittampalam, and Edward Aftandilian. 2024. Measuring GitHub Copilot's Impact on Productivity. *Commun. ACM* 67, 3 (Feb. 2024), 54–63. doi:10.1145/3633453

A Ban Lift Analysis

To further examine the causal relationship and strengthen our interpretation, we analyze the period following the ban's lift on April 28, 2023. This analysis serves as a complementary test of our circumvention hypothesis: if Italian developers experienced a genuine productivity decline during the ban (due to inability to access ChatGPT), we would expect to observe a positive rebound effect when access was officially restored. Conversely, if developers successfully circumvented the ban and maintained access throughout the restriction period, the official lifting should have no effect—they would already be using ChatGPT regardless of its legal status. This logic provides a powerful test of whether the ban was effectively enforced or circumvented.

Table 4 presents difference-in-differences estimates comparing coding activity before and after the ban was lifted, using the same time windows as our main analysis. The coefficient on "Italy × Post-Lift" represents whether developers in Italy experienced a rebound in coding activity relative to the control countries once ChatGPT access was officially restored.

We find no statistically significant effects in any of the specifications. For the two-week post-lift period, the treatment effects are -0.0321 (SE: 0.0494) for log additions and -0.0044 (SE: 0.0502) for log deletions. For the four-weekday period, the effects are -0.0158 (SE: 0.0541) and -0.0585 (SE: 0.0547), respectively. All coefficients are small in magnitude, not statistically significant, and show no consistent pattern of positive rebound.

The absence of any observable increase in coding activity following the ban's lift is consistent with our main interpretation: developers had already restored access through circumvention (primarily VPNs, as documented in Appendix B) weeks before the official lifting. If they were already using ChatGPT throughout the ban period, the legal change on April 28 would have no marginal impact on their actual access or productivity, which is precisely what we

Table 4. Difference-in-Differences Results: Post-Ban Lift (Absorbed Fixed Effects)

	Log Additions		**Log Deletions**	
	2 Weeks	4 Weekdays	2 Weeks	4 Weekdays
Italy × Post-Lift	-0.0321	-0.0158	-0.0044	-0.0585
	(0.0494)	(0.0541)	(0.0502)	(0.0547)
Observations	209142	169574	209142	169574
R^2	0.436	0.457	0.430	0.451

Notes: Absorbed fixed effects for users and dates. Robust standard errors are clustered on the user-level in parentheses. *** p<0.01, ** p<0.05, * p<0.1

observe. This null result, combined with the null finding during the ban period itself, provides mutually reinforcing evidence that the treatment was effectively porous due to widespread circumvention.

B Ban Circumvention Analysis

This appendix presents the difference-in-differences analysis of search volume for circumvention tools—specifically VPN services and Tor Browser—in Italy (treatment group) compared to Austria and Germany (control groups) before and after the ban implementation on April 1st, 2023. The data was obtained from Google Trends,[4] which provides normalized search interest values representing the share of web searches for a given term relative to all searches on Google over time. The values are normalized on a scale of 0–100, where 100 represents peak popularity for the term during the selected time period.

Figure 1 shows the VPN search volume over time for Italy, Austria, and Germany. The vertical dashed line indicates the ban start date (April 1st, 2023). Prior to the ban, Italy showed consistently lower baseline search interest in VPNs compared to the control countries, with average values around 21 points versus 33–35 points for Austria and Germany. Following the ban implementation, Italy experienced a dramatic spike in VPN search interest, reaching a peak of 100 (the maximum normalized value) on April 1st, and maintaining elevated levels throughout the post-ban period with an average of 51 points.

Figure 2 compares the average VPN search volume before and after the ban implementation across the three countries. The visualization clearly demonstrates the substantial increase in VPN search interest in Italy following the ban, while the control countries (Austria and Germany) showed minimal changes, consistent with the parallel trends assumption required for difference-in-differences estimation.

Table 5 presents the difference-in-differences regression results for both VPN and Tor Browser searches. The dependent variable in each model is search volume (normalized Google Trends index). The coefficient on "Italy × Post-Ban" represents the treatment effect for each circumvention tool. For VPN searches, the coefficient indicates that searches in Italy increased by 28.6 points (on the 0–100 scale) more than in the control countries after the ban implementation, an effect that is statistically significant at the 1% level ($p < 0.001$). For Tor Browser searches, the treatment effect is 20.2 points, statistically significant at the 5% level ($p < 0.05$). These results suggest a substantial increase in interest in both VPN services and Tor Browser as means of circumventing the ban. The magnitude of the VPN effect represents more than a doubling of baseline search interest in Italy, consistent with users seeking tools to bypass geographic restrictions.

[4]https://trends.google.com/

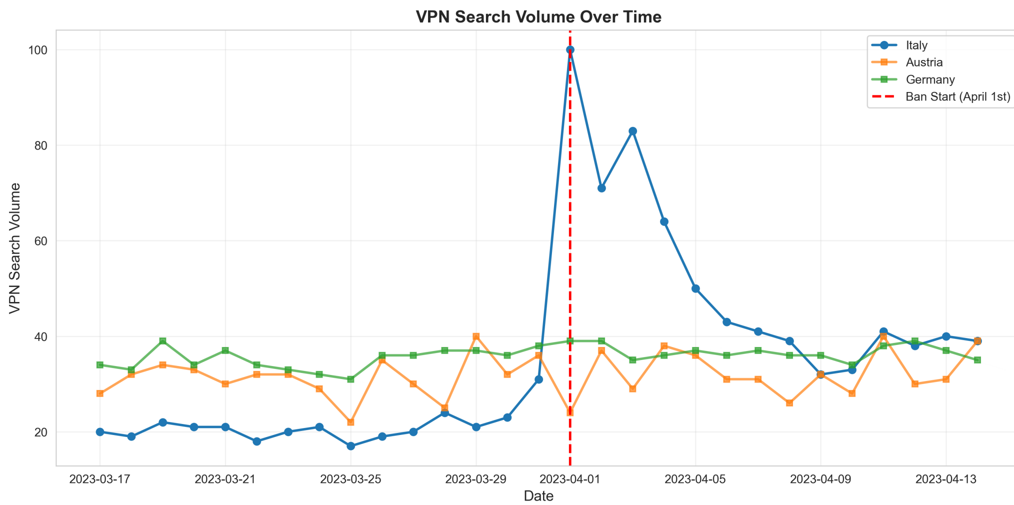

Fig. 1. VPN Search Volume Over Time

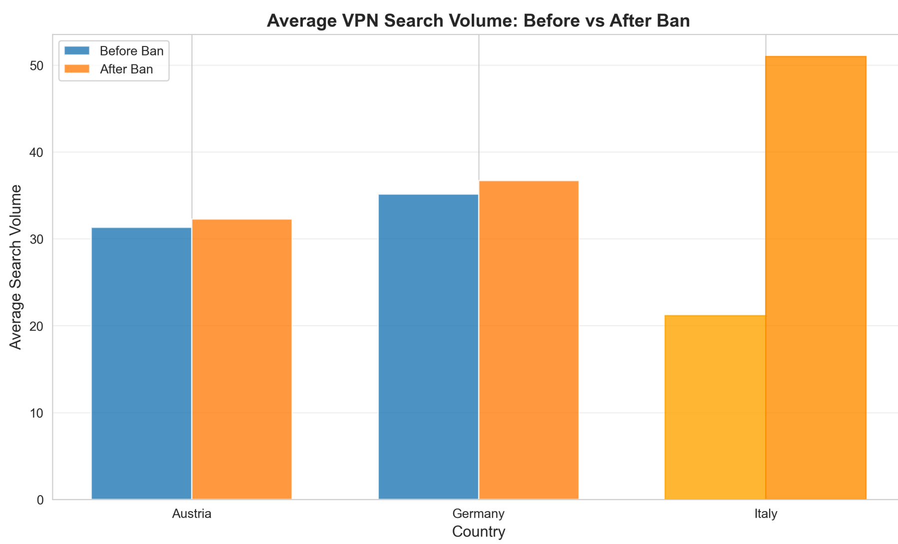

Fig. 2. Average VPN Search Volume: Before vs After Ban

Table 5. Difference-in-Differences Regression Results

	Dependent variable: search_volume	
	VPN	Tor Browser
Italy	-12.100***	-10.167
	(2.823)	(6.221)
Post-Ban	1.267	-0.710
	(2.346)	(5.169)
Italy × Post-Ban	28.600***	20.167**
	(4.063)	(8.953)
Constant	33.233***	17.567***
	(1.630)	(3.592)
Observations	87	87
R^2	0.497	0.079
Adjusted R^2	0.479	0.046
Residual Std. Error	8.928 (df=83)	19.672 (df=83)
F Statistic	27.344*** (df=3; 83)	2.371* (df=3; 83)
Note:		*p<0.1; **p<0.05; ***p<0.01